PSC 85509: Applied Quantitative Research II

Instructor: Thomas Leavitt

Official Course Description

This course is the third in the Political Science program's quantitative methods sequence, following PSC 85507 (Quantitative Analysis I: Regression Analysis). The course is intended for students who want to develop the skills to use quantitative methods in their own research and to think systematically about empirical data in political science. The course begins with a review of the foundations of statistical modeling, estimation, and inference, including ordinary least squares (OLS) and maximum likelihood, and then examines how to apply these tools in common empirical settings. Topics include methods for panel, multilevel (hierarchical), event history (survival), and spatial data, as well as the use of these methods in causal inference designs, such as regression discontinuity, difference-in-differences, and instrumental variables.

Students will work with real datasets, replicate and extend published empirical studies, and produce an original empirical research paper applying these methods to a substantive question in political science. The course uses both `R` and `Stata` for statistical analysis. By the end of the semester, students will be able to identify appropriate quantitative methods for different types of data, implement these methods using statistical software, and apply them to conduct and evaluate research in political science.

Course Organization

The course assumes the material of PSC 85507. From that course you should bring descriptive statistics and the normal distribution; bivariate and multiple regression fit by least squares; hypothesis tests, p-values, and confidence intervals; dummy variables, interactions, polynomials, and logarithms as ways of changing a regression's functional form; the omitted-variable-bias formula; heteroskedasticity, autocorrelation, multicollinearity, and influential observations; logit and probit, their coefficients converted to predicted probabilities, odds ratios, and average marginal effects; and `Stata`. This course assumes no prior experience with `R`.

The course has two parts. Part I (Weeks 1–5) builds the framework, and three of its five weeks — Weeks 2, 4, and 5 — are also anchored in a published application. Part II (Weeks 6–13) applies the framework to causal questions and to dependent and incomplete samples, and anchors every week in a published application. Week 14 adds final-paper presentations.

The course gives you a framework that outlasts any particular method. New methods keep arriving: The omitted-variable-bias sensitivity analysis taught in Week 8, the difference-in-differences estimators surveyed in Week 11, and double machine learning (Chernozhukov et al., 2018; Kennedy, 2024), which this course does not cover, all emerged within the last decade. With the fundamentals in hand, you can teach yourself whatever comes next.

Part I (Weeks 1–5): The framework

Week 1: Estimands, estimators, and the plug-in principle

We begin not with a method but a question: *What is the quantity you want to know?* That quantity is the *estimand*; a recipe turning data into a guess of the estimand is an estimator, and the guess the estimator returns on a particular sample is an estimate. An estimand is a feature of the population distribution P from which each observation is drawn, written $\psi(P)$.

We never observe P, only a sample of n observations, from which we form the *empirical distribution* $\hat{P}$ putting probability $1/n$ on each observation. The *plug-in principle* says: To estimate $\psi(P)$, compute the same function on $\hat{P}$. For a population mean, $\psi(\hat{P})$ is the familiar sample mean, and the gap between $\psi(\hat{P})$ and $\psi(P)$ is what Weeks 3 and 4 quantify. Nearly every estimator in this course is a plug-in estimator, and the first question we ask of any procedure is: *Which population quantity is this procedure the plug-in for?*

Week 2: The CEF, the BLP, and OLS as the BLP's plug-in

Week 2 takes the most familiar tool in applied work — OLS, the recipe that picks the coefficients of a linear function of predictors to minimize the average squared residual in the sample — and reframes it around two features of the population distribution, each a $\psi(P)$. The first, the *conditional expectation function* (CEF), is the population average of the outcome Y at each value of the predictors X. The second, the *best linear predictor* (BLP), is the linear function of X that predicts Y with the smallest average squared error under P. When the CEF is linear, the two coincide; otherwise the CEF and the BLP are genuinely different quantities. *Under random sampling, OLS converges to the best linear predictor of Y given X in the population* — whether or not the CEF is linear, whether or not the error (the deviation of Y from the BLP) is normal, and whether or not that error has constant variance. We therefore treat OLS not as a model but as the plug-in estimator of the BLP, since minimizing the average squared residual in the sample is the BLP's own minimization carried out under $\hat{P}$ in place of P. Which BLP OLS converges to depends on which predictors enter the regression. Each set of predictors defines its own BLP in the population, so the BLP coefficients on a given set of predictors generally differ according to which other predictors the regression includes, and the omitted-variable-bias formula computes that difference exactly, without any model of the population.

Week 3: Consistency — the gap between estimator and estimand, and how fast that gap shrinks

The foundations that Weeks 3–4 develop are *asymptotic* — exact only in the limit as n increases indefinitely, approximate at the n we actually have — so the inferences built on those foundations hold only approximately in any actual sample. Week 3 establishes *consistency*. The Weak Law of Large Numbers makes the sample mean $\bar{X}$ converge in probability to the population mean

μ, meaning that for any margin, however small, the probability that $\bar{X}$ falls within that margin of μ tends to one as n grows. The CONTINUOUS MAPPING THEOREM carries that convergence through continuous functions. Nearly every estimand in the course is a continuous function of population means, and the plug-in estimator of that estimand is the same function of the corresponding sample means, so each plug-in estimator $\psi(\hat{P})$ converges to its estimand $\psi(P)$. The *rate of convergence* says how fast each estimator approaches its estimand. The typical magnitude of the estimator's error — the gap between $\psi(\hat{P})$ and $\psi(P)$ — shrinks in proportion to $1/\sqrt{n}$, so a sample four times as large halves that magnitude.

Week 4: Agnostic inference — the distribution of the estimator's error, with no model of the population

To enable hypothesis tests — calculations of how unusual an estimate $\psi(\hat{P})$ would be if $\psi(P)$ took a hypothesized value — we need a probability distribution for the error $\psi(\hat{P}) - \psi(P)$. In the absence of a model of the population, Week 4 appeals to the CENTRAL LIMIT THEOREM and associated theory, which make that error approximately normal in large samples when $\psi(P)$ is a population mean, so that $\psi(\hat{P})$ is a sample mean. SLUTSKY'S THEOREM then lets us substitute a consistent estimator — computed from the sample — for the unknown standard deviation of that distribution, the *standard error*, leaving the normal approximation intact. The DELTA METHOD carries that normal approximation from the error in a sample mean to the error in a differentiable function of sample means, which estimates the corresponding function of population means. Nearly every estimand in this course takes that form. Alternatively, the *bootstrap* approximates the same error distribution without invoking normality, repeatedly drawing samples of size n from $\hat{P}$ itself — that is, resampling the observed data with replacement.

The template of Weeks 3–4 recurs for every method in the course: *Choose the estimand first, confirm the recipe tracks that estimand, and get the distribution of the estimator's error from the normal approximation with an estimator of the standard error or from the bootstrap.* The few estimators that depart from the template do so instructively, and Part II flags the departures that matter. Nothing that follows assumes the population is normal or that the fitted equation matches the CEF. Nor do we assume the errors have constant variance, which the classical formula for the standard error requires, so heteroskedasticity-robust ("sandwich") estimators of the standard error are our default.

Week 5: Parametric models as plug-ins — what maximum likelihood targets when the model is false

A model is *parametric* when the outcome's distribution at each value of the predictors is described by a finite list of parameters whose length does not depend on the sample size. Logistic regression, probit regression, and their generalized-linear relatives are all parametric, each pairing a distribution for the outcome with a linear function of the predictors. *Maximum likelihood* fits such models by solving the sample version of a population problem, taking the parameter values that maximize the sample's average *log-likelihood*, which quantifies how probable the observed data are under the model at those values. Just as the BLP is the linear function of X with the smallest average squared error under P, the population version of that problem asks for the parameter values with the largest average log-likelihood under P. Under conditions we state — most importantly that those parameter values are unique — the sample maximizer converges to them, even if the model defining the likelihood is incorrect. The same convergence holds for the quantities we compute from the fitted parameters, each of which has a population target of its own. A difference in predicted probabilities, for instance, has a target determined by the two values you contrast and by where you hold the remaining predictors. The *sandwich* variance estimator supplies standard errors for the fitted parameters without assuming the model is correct, and Week 4's delta method extends those standard errors to differentiable functions of the parameters, a difference in predicted probabilities among them. Hypothesis tests built this way are valid in large samples for these population targets, even when the model is false.

Part II (Weeks 6–13): From causal questions to dependent and incomplete samples

So far we have described populations and predicted outcomes. *Causal* inference adds a step, the specification of a causal estimand: a treatment effect, or how outcomes would differ if treatment were different. Suppose A equals one for the treated and zero for the untreated — the two *treatment arms* — and Y is the outcome. A population target $\psi(P)$ might then contrast the distribution of outcomes among the treated with the distribution among the untreated, comparing two parts of one population as it actually is. The analogous causal estimand contrasts the distribution of outcomes the whole population would have were everyone treated with the distribution that same population would have were no one treated.

A causal estimand is thus a feature of the distributions the population would have under each treatment, not of P, the distribution the population actually has. Even if you knew P exactly, the causal estimand would remain undetermined without further assumptions. Such *identification assumptions* state the conditions under which $\psi(P)$ coincides with the causal estimand, so that your statistical tools warrant causal conclusions. Of every method in Part II, we ask not only *what feature the recipe targets in the distribution P that you actually sample from*, but also *under what assumptions that feature is a causal estimand*.

Weeks 6–8: Identification from observed covariates — causal estimands, conditional ignorability, and sensitivity analysis

Week 6 opens with the potential outcomes framework, which defines causal effects in terms of the outcomes each unit would have under each treatment, and with the identifying assumption we rely on most, *conditional ignorability*. To state that assumption, fix a predictor profile — a single set of values of the predictors — at which both treatments occur, as they must at every profile for what follows, so that P supplies a distribution of outcomes among the treated and a distribution among the untreated. Conditional ignorability says that, at any such profile, the distribution of outcomes among the untreated is also the distribution the treated would have had were they untreated, and vice versa. Causal identification follows. The distribution of outcomes among the treated at each profile, averaged over profiles in proportion to their shares of the population, is the distribution of outcomes the whole population would have were everyone treated, and likewise for the untreated. The causal estimand is typically the difference between the means of those two counterfactual distributions, the *average treatment effect*.

Week 7 turns from identification to estimation. Two routes construct plug-in estimators of $\psi(P)$, which conditional ignorability makes the average treatment effect. One route models the outcome, predicting each unit's outcome under each treatment and averaging the predictions. The other models the treatment — the *propensity score*, the probability that a unit with a given predictor profile receives treatment — and weights each unit by the inverse of the estimated probability of the treatment that unit received. The outcome model is correct only when its predictions converge to the CEF, which for OLS means the BLP equals the CEF; the treatment model is correct only when its predicted probabilities converge to the propensity score. Combining the two models yields a *doubly robust* estimator, consistent for $\psi(P)$ when either model is correct.

Conditional ignorability is itself untestable, since the data cannot reveal what an unobserved predictor would have shown. Regression still estimates $\psi(P)$ whether or not the assumption holds; what fails is the causal reading. Week 8's sensitivity analysis therefore uses Week 2's omitted-variable-bias formula, asking what associations an unmeasured confounder would need to have with the treatment and the outcome to overturn your causal conclusion.

Weeks 9–11: Identification from instruments, thresholds, and timing — exclusion restrictions, continuity at the cutoff, and parallel trends

Weeks 9–11 turn to identifying assumptions that do not rest on conditional ignorability. These assumptions attach instead to an instrument that moves treatment without affecting the outcome directly (*the exclusion restriction*, Week 9), an eligibility threshold that renders units just above and just below a cutoff comparable (*continuity at the cutoff*, Week 10), and the timing of a policy's adoption, with never-adopting units tracing the path the adopters would otherwise have followed (*parallel trends*, Week 11). Each design has a workhorse recipe — two-stage least squares, the local polynomial fit, and the two-way fixed-effects regression, respectively — and of each we ask the two questions of Part II: which $\psi(P)$ the recipe converges to, and what the design's assumption contributes to making that $\psi(P)$ the effect the design advertises. Two of these recipes depart from the template of Weeks 3–4, both in what they target and in the distribution of their error: the local polynomial fit (Week 10), and the two-way fixed-effects regression when units adopt treatment at different times (*staggered adoption*, Week 11). In each case we ask what estimation and inference must do instead.

Weeks 12–13: Dependent and incomplete samples — partial pooling, censoring, and spatial dependence

Weeks 12–13 confront the premise the course has so far relied on quietly: that observations reach us as independent, fully observed draws from P. Week 12 covers multilevel models, which pool information across groups (*partial pooling*) — voters within districts, students within schools — whose members are not independent draws, and event-history models, which follow units toward events that may not occur before the observation window closes (*censoring*). Week 13 covers spatial models, which let one unit's outcome depend on the outcomes of neighboring units (*spatial dependence*). In each case the recipes of Weeks 3–4 no longer apply as stated, since the guarantees behind those recipes assume independent draws and complete observation. We ask in each case which premise failed and what estimation and inference must do instead.

Course Learning Goals

By the end of this course, students should be able to do the following.

1. Articulate what an estimator targets in a population, and reason carefully about whether the estimator reaches that target under assumptions you are willing to defend.
2. Distinguish the *statistical* question from the *causal* question in any analysis, your own or a published study's, and evaluate the identifying assumptions that let a statistical quantity answer the causal question.
3. Implement the core methods of quantitative political science research in `R` or `Stata` (course code is provided in both): OLS with sandwich and cluster-robust inference, maximum-likelihood and other plug-in estimators, propensity-score weighting, doubly robust estimation, sensitivity analysis for unmeasured confounding, instrumental variables, regression discontinuity, difference-in-differences, multilevel models, event history analysis, and spatial models.
4. Read and critically evaluate published quantitative political science research, including identifying threats to the validity of its conclusions and proposing principled extensions.
5. Replicate and extend a published empirical study, communicating your analysis through reproducible code, well-formatted tables and figures, and clear written interpretation.
6. Most importantly, learn a new method on your own — from a methods paper, a package vignette, or a replication archive — by asking what the method targets, under what assumptions, and how to quantify the error of its estimator.

Course Assignments

Problem sets (4)

Each student will complete four problem sets, each due at the start of class in the week marked in the schedule below. The four sets follow the course's arc: Problem set 1 covers the plug-in framework, OLS, and consistency (Weeks 1–3); problem set 2, agnostic inference and maximum likelihood (Weeks 4–5); problem set 3, causal identification, weighting, and sensitivity analysis (Weeks 6–8); and problem set 4, instrumental variables, regression discontinuity, and difference-in-differences (Weeks 9–11).

Problem sets will combine analytical questions and replication exercises in `R` and/or `Stata`. Problem sets 3 and 4 add an "Application of Methods" component in which student groups select a published political science article that uses that problem set's methods and evaluate the article's use of those methods. Each group should select a different article; groups coordinate their choices in a shared spreadsheet, first come, first served. Articles covered in lecture or listed in the slide references are not eligible for the Application of Methods component.

On any problem set's due date, I may ask you to complete a short *check of understanding* tied to the problem set just submitted: a brief handwritten answer, *no laptop*, to one of its questions, or a short account of your own submitted code.

Topic-week replication and co-teaching (Weeks 7–13)

Beginning with Week 7 (the first week after the causal-inference foundations of Week 6), each topic week features a **designated student** who will replicate the study listed under **Application** in that week's readings below and co-teach the application portion of the session with me. The role is substantial: You walk the class through the data, the analysis, the identification strategy, the diagnostics, and the substantive interpretation, and then lead the discussion. Your job is to connect the week's conceptual material to a worked empirical example.

There are seven topic weeks (Weeks 7–13), and students claim them at the start of the term, first come, first served. If the class is smaller than seven, I will lead the application in any week that goes unclaimed; if the class is larger, students will work in pairs or small groups to co-lead a week. Choose a topic aligned with your interests — ideally one connected to your final paper topic, so that the replication contributes directly to the final paper. You may also propose an application other than the one listed (subject to my approval), provided the alternative is published, uses the appropriate method on the appropriate data structure, and has available replication data.

You will submit a written replication report — your full code, in `R` or `Stata`, plus a 3–5 page memo describing the design, analysis, and any extensions or critiques — one day before your presentation. The presentation itself should run roughly 20–25 minutes, leaving the rest of the session for instructor-led content and full-class discussion.

Final research paper

Each student will complete an original empirical research paper applying the course's methods to a substantive question of their choosing. The paper should be a polished, professional document suitable as a writing sample for the job market, a draft for journal submission, or a policy report, depending on each student's career objectives.

I strongly encourage students who completed a research paper or replication project in PSC 85507 (or another course) to continue developing that work in PSC 85509, ideally toward a publishable piece. The methods you will learn this term will let you take a paper you wrote with the toolkit of PSC 85507 and deepen the paper's identification strategy, extend the paper to a different data structure, or reframe the paper as a more credible empirical study. The project milestones described below are intended to give you time to develop the paper substantively.

Project milestones

- **Research proposal** (due Week 6): a 2–3 page document outlining the research question, motivating literature, proposed methodology, and intended data source(s).

- **Draft of final paper** (due at the end of Week 13): a complete first draft of the paper, shared with your discussant on submission.

- **In-class presentation** (Week 14): a 15-minute talk on the paper, with slides. Each student also serves as the discussant for one classmate's paper, having read the classmate's draft, and delivers 5 minutes of remarks, also with slides. Both slide decks are due by the end of the day before the presentation.

- **Final paper** (due the last day of finals week): the polished final version, incorporating feedback from the draft and presentation.

Grades

Final grades weight the course assessments as follows.

- **28%**: four problem sets (**7%** each).
- **20%**: the topic-week replication and presentation.
- **5%**: the research paper proposal.
- **5%**: the draft of the final paper.
- **30%**: the final paper.
- **12%**: the in-class final-paper presentation, attendance, and participation (including service as a discussant for classmates' presentations).

Course Readings

This course is ZTC ("Zero Textbook Cost"). I will post all required readings on the course website, and the reading list draws primarily from the following books, all freely available online or through the library. I will also upload the dataset for each week's application to the course website, so no assignment requires you to track down data.

1. Hansen, Bruce E. (2022). *Probability and Statistics for Economists*. Princeton, NJ: Princeton University Press. (Cited below as "Hansen Vol I.")
2. Hansen, Bruce E. (2022). *Econometrics*. Princeton, NJ: Princeton University Press. (Cited below as "Hansen Vol II.")
3. Aronow, Peter M., and Benjamin T. Miller. (2019). *Foundations of Agnostic Statistics*. New York, NY: Cambridge University Press.
4. Gelman, Andrew, Jennifer Hill, and Aki Vehtari. (2021). *Regression and Other Stories*. New York, NY: Cambridge University Press.
5. Hernán, Miguel A., and James M. Robins. (2020). *Causal Inference: What If*. Boca Raton, FL: Chapman & Hall/CRC. Available online at `https://miguelhernan.org/whatifbook`. (Page references follow the online edition dated November 21, 2025.)
6. Cattaneo, Matias D., Nicolás Idrobo, and Rocío Titiunik. (2020). *A Practical Introduction to Regression Discontinuity Designs: Foundations*. New York, NY: Cambridge University Press. (*Elements in Quantitative and Computational Methods for the Social Sciences.*)
7. Gelman, Andrew, and Jennifer Hill. (2007). *Data Analysis Using Regression and Multilevel/Hierarchical Models*. New York, NY: Cambridge University Press.
8. Roback, Paul, and Julie Legler. (2021). *Beyond Multiple Linear Regression: Applied Generalized Linear Models and Multilevel Models in R*. Boca Raton, FL: Chapman & Hall/CRC. Available online at `https://bookdown.org/roback/bookdown-BeyondMLR/`.

9. Box-Steffensmeier, Janet M., and Bradford S. Jones. (2004). *Event History Modeling: A Guide for Social Scientists*. New York, NY: Cambridge University Press.

10. Ward, Michael D., and Kristian Skrede Gleditsch. (2007). *An Introduction to Spatial Regression Models in the Social Sciences*. Unpublished manuscript, June 15, 2007. (Week 13's page references follow this manuscript.)

In addition to texts focused on statistical and causal inference methods, virtually every week's reading list includes one or more articles applying those methods to substantive empirical questions. We will draw applications from American politics, comparative politics, and international relations.

Course Standards

Electronics

To the extent possible, **please bring a personal laptop to each class** for class-related purposes, mostly in-class exercises using statistical software. However, please do not use your cell phones or other electronic devices during class. If a cell phone or another device is important for your participation in the course, please let me know; I will adjust the policy for the entire class rather than single anyone out.

Attendance, Participation, and Collaboration

Attendance and active participation are extremely important to learning from this course. I strongly encourage you to collaborate with other students on problem sets, and the "Application of Methods" components are explicitly group-based. However, all written submissions should be in your own words.

If you need to miss class, please do **not** contact me ahead of time. A missed class affects the participation component of your grade the same way whether or not you contact me beforehand. Please contact me only if you have to miss a class in which you are scheduled to present.

Incomplete and Late Assignments

If you think that you will be unable to complete an assignment by its due date because of extenuating circumstances, please **do** let me know ahead of time. I may change the assignment's deadline for the entire class. If you turn in the final paper late without my prior consent, I will deduct one-third of a letter grade per day. Because I typically review problem sets in class, I will not accept late problem sets for credit.

Use of Artificial Intelligence (AI) Tools

Because this course is fundamentally about quantitative reasoning and statistical computing, I am not forbidding these tools that are reshaping how such work is done. You are welcome — encouraged, even — to use large language models (ChatGPT, Claude, Copilot, and the like) to debug your `R` or `Stata` code, decipher cryptic error messages, brainstorm questions worth asking of the data, and draft the routine syntax you would otherwise type out by rote.

A single well-posed prompt already takes you a long way. A scripted pipeline takes you further: It sweeps through thousands of records, merges them with other sources, and logs every step, which ad hoc prompting cannot do at any scale you can audit or reproduce.

That latitude comes with one non-negotiable condition: You own everything you submit.

- **The output is yours.** "The model hallucinated" and "that is just what the chatbot gave me" are never valid defenses. If your code quietly returns the wrong answer, or your paper cites a study that was never written, the error is yours and no one else's. Verify everything these tools produce before your name goes on anything you used the tools for.
- **You must be able to explain it.** You may use AI to help write code, but you are answerable for every line you hand in. If I ask you in class or in office hours to walk me through a particular function or loop in your work and you cannot say how the function or loop behaves or why it is there, that portion of the assignment may receive no credit. More important still, I want you to ask at every step what could be wrong with a given result — what sanity check, cross-tabulation, or quick plot would tell you whether you are truly on the right track. Any check of understanding on a problem set's due date asks whether you can explain your own submission, unaided. The ability to do so is what points to sound use of AI tools.
- **Think before you prompt.** These tools are excellent at carrying out a task you have already framed and poor at supplying the insight that framed it. Do not surrender your judgment to them; use them to move faster on questions that are already your own, not to do your thinking for you.
- **Disclosure.** Any assignment on which you relied on AI tools must include a brief disclosure statement. As journals ask of their authors, it should (i) name the specific tool(s) and the version or model you used (for example, the product name together with its model or version identifier), (ii) state what you used each one for — which tasks, and which parts of the work — and (iii) confirm that you reviewed and verified the output and take full responsibility for it. As in those venues, an AI tool is never a co-author and never answerable for the work; that responsibility is yours alone.

The course's academic-integrity standards apply to AI-assisted work exactly as they do to everything else, and I treat undisclosed use of AI as a breach of them. These tools also routinely misattribute or invent their sources, so catching and correcting such errors before you submit is part of the same responsibility.

Academic Integrity

I will not tolerate violations of academic integrity under any circumstances. Please familiarize yourself with the CUNY Graduate Center Academic Integrity Policy. If you are having difficulty with an assignment, do not attempt to present another author's work as your own; please come talk to me instead.

Software and Computing

This course uses both R and Stata, and the bilingualism is my job, not yours: **All code I provide — in lectures, problem sets, and replication exercises — will be available in both languages**, so that the course accommodates every student's choice. **There is no bilingual requirement for any student work.** Submit every assignment — problem sets, the replication, the final paper — in whichever language you prefer. PSC 85507 is taught in Stata; although nothing you submit has to be in R, a goal of this course is for you to leave equally comfortable in R.

Why R is worth the investment

If you have not yet used R seriously, the start of this course will require an investment of time. That investment pays off, for several reasons.

- **Frontier methods are R-first.** New causal-inference and machine-learning methods typically appear in R years before they reach Stata, if they reach it at all. The did, DoubleML, HonestDiD, and sensemakr packages reached Stata only when other authors later reimplemented the R originals; causal forests (grf) waited years for a Stata counterpart; and at the time of this writing, kernel balancing (kbal) has no Stata implementation at all. Working at the methodological frontier usually means working in R.
- **Free and open source.** R runs on any platform, costs nothing, and requires no license and no institutional affiliation. That accessibility matters for your collaborators, for replication archives, and for the work you do after graduate school.
- **The standard in political science methodology.** Replication archives for *Political Analysis*, the *American Journal of Political Science*, the *Journal of Politics*, and similar venues now overwhelmingly ship R code. You need R to read recent methods papers in political science.
- **Reproducible research workflows.** R Markdown and Quarto integrate code, output, citations, and prose into a single document, so you can write a polished, fully reproducible paper or report without ever leaving that document.
- **Visualization.** The ggplot2 package gives you finer control over a figure than Stata's graphics commands do. Political science journals increasingly publish figures that authors build in ggplot2.
- **Job market.** Data-science, government-analyst, and industry positions often expect R or Python (or both). Stata alone narrows your opportunities outside academic econometrics. Of the software skills you can build in this course, R repays you the most.

Installing the software

To download R, please go to https://cloud.r-project.org and follow the installation instructions. After installing R, please download and install RStudio from https://posit.co/download/rstudio-desktop/. CUNY provides access to Stata, which you can download for your personal computer; Stata is also installed on lab computers at the Graduate Center.

A valuable free resource for learning R is Wickham et al. (2023), available at https://r4ds.hadley.nz/.

Course Schedule and Readings

The CUNY Graduate Center's academic calendar is available via the Office of the Registrar here.

Within a week, anything **due** comes first, marked ⋆ and set in **blue**; anything distributed that week follows, marked ∘.

Part I (Weeks 1–5): The framework

Week 1: Estimands, estimators, and the plug-in principle

Required readings

Theory

- Aronow and Miller (2019) §§2.1.1–2.1.3 (pp. 45–59), §3.1 (pp. 91–96), §3.3 (pp. 116–120)

Recommended readings

- Hansen Vol I §§6.4–6.11 (pp. 130–137)
- Wickham et al. (2023): available here

Week 2: The CEF, the BLP, and OLS as the BLP's plug-in

∘ Problem set 1 distributed

Required readings

Theory

- Aronow and Miller (2019) §§2.2.3–2.2.5 (pp. 67–84), §4.1 (pp. 143–151) — read §§4.1.1–4.1.2 (pp. 144–145) before §4.1.3 (p. 147)

Theory (omitted variable bias)

- Hansen Vol II §2.24 (pp. 45–46)

Application

- Mincer (1974), used throughout Hansen Vol II Chs. 2–4

Recommended readings

- Hansen Vol II §§2.3–2.18 (pp. 16–40)

Week 3: Consistency — the gap between estimator and estimand, and how fast that gap shrinks

Required readings

Theory

- Aronow and Miller (2019) §3.2.1 (pp. 96–102, convergence in probability, the continuous mapping theorem, the weak law of large numbers), §3.2.2 (pp. 102–105, bias, consistency, mean squared error), §3.2.3 (pp. 105–108, the plug-in sample variance, the unbiased sample variance)

- Aronow and Miller (2019) §3.3.1 (pp. 120–121, the plug-in regularity conditions)

Recommended readings

- Hansen Vol I §7.3 (p. 149), §7.5 (p. 152), §7.10 (p. 155)

Week 4: Agnostic inference — the distribution of the estimator's error, with no model of the population

⋆ **Problem set 1 due**

Required readings

Theory (asymptotic normality)

- Aronow and Miller (2019) §3.2.4 (pp. 108–111, convergence in distribution, the central limit theorem, Slutsky's theorem), §3.2.5 (pp. 111–114, asymptotic normality)
- Hansen Vol I §8.9 (p. 172, the delta method)

Theory (sandwich)

- Aronow and Miller (2019) §3.2.6 (pp. 114–120), §3.4 (pp. 124–135), §3.5 (pp. 135–141), §4.2 (pp. 151–156)
- Hansen Vol II §§4.15–4.16 (pp. 111–114) and §§4.22–4.23 (pp. 123–128, cluster-robust)

Theory (bootstrap)

- Aronow and Miller (2019) §3.4.3 (the bootstrap, within §3.4 above)
- Hansen Vol II §10.29 (p. 299) and §10.30 (p. 301, wild and cluster bootstrap)

Application

- Gilens and Page (2014)

Recommended readings

- Hansen Vol I §8.11 (p. 173, the asymptotic distribution of a plug-in estimator), §8.2 (p. 165), §8.6 (p. 170)
- Hansen Vol II §4.24 (pp. 128–130, at what level to cluster), §7.3 (p. 164, asymptotic normality of OLS)
- Cameron and Miller (2015)
- Cameron et al. (2008)
- Efron and Tibshirani (1993, Chs. 1–7)
- Gelman et al. (2021, §§4.4–4.5, pp. 57–62, Type S and Type M error, the significance filter, forking paths)

Week 5: Parametric models as plug-ins — what maximum likelihood targets when the model is false

- ○ Problem set 2 distributed

Required readings

Theory

- Aronow and Miller (2019) §5.1 (pp. 178–185), §5.2 (pp. 185–203), especially §5.2.3 (p. 194, maximum likelihood under misspecification), §5.2.4 (p. 197, the maximum-likelihood plug-in estimator), §5.2.7 (p. 202)
- Gelman et al. (2021, §14.4, pp. 249–252, average predictive comparisons)

Application

- Fearon and Laitin (2003)

Recommended readings

- Hansen Vol II Ch. 22 (pp. 769–777, M-estimation as a general class)

Part II (Weeks 6–13): From causal questions to dependent and incomplete samples

Week 6: Causal estimands, identification, and conditional ignorability

- ⋆ **Research proposal due**

Required readings

Theory

- Aronow and Miller (2019) §7.1 (pp. 236–256), especially §7.1.2 (p. 238, missing data), §7.1.5 (p. 247, ignorability), §7.1.6 (p. 250, the propensity score), §7.1.7 (p. 252, post-treatment variables)
- Hernán and Robins (2020, Chs. 1–3, pp. 3–46)

Application

- Hainmueller and Hangartner (2013)

Recommended readings

- Imbens and Rubin (2015, Chs. 1–3)
- Gelman et al. (2021, Ch. 18, pp. 339–362; identification at p. 351, the stable unit treatment value assumption (SUTVA) at pp. 352–354)

Week 7: Estimation after identification — outcome models, propensity scores, and doubly robust estimation

- ⋆ **Problem set 2 due**
- ○ Problem set 3 distributed

Required readings

Theory (outcome models)

- Aronow and Miller (2019) §7.2.1 (p. 256), §7.2.2 (p. 258), §7.2.3 (p. 261, maximum-likelihood plug-in estimation)

Theory (weighting)

- Aronow and Miller (2019) §7.2.5 (p. 264), §7.2.6 (pp. 264–267), §7.2.7 (pp. 267–270, doubly robust estimators), §7.2.8 (pp. 270–271), §7.3 (pp. 271–276, overlap and positivity)

Application

- Blattman (2009)

Recommended readings

- Gelman et al. (2021, Ch. 20, pp. 399–413, the propensity-score workflow)

Week 8: Sensitivity analysis — how strong unmeasured confounding would have to be

Required readings

Theory

- Cinelli and Hazlett (2020, §3.1, pp. 43–45, omitted variable bias; §4.3, pp. 50–52, the robustness value; §4.4, pp. 52–54, benchmarking)
- Hansen Vol II §2.24 (pp. 45–46), revisited from Week 2

Application

- Hazlett (2020)

Recommended readings

- Gelman et al. (2021, Ch. 20, pp. 383–394, omitted variable bias; the algebra at p. 385)
- Hosman et al. (2010): the earlier omitted-variable-bias approach to sensitivity analysis that Cinelli and Hazlett (2020) §6.1.2 compares itself against

Week 9: Instrumental variables — the effect for compliers

⋆ **Problem set 3 due**

Required readings

Theory

- Angrist et al. (1996)
- Hansen Vol II §§12.1–12.12 (pp. 337–352), §12.34 (p. 388, the local average treatment effect), §12.36 (p. 392), §12.38 (p. 399, weak instruments)

Application

- Acemoglu et al. (2001)

Recommended readings

- Hernán and Robins (2020, Ch. 16, pp. 209–226)
- Sovey and Green (2011)
- Gelman et al. (2021, §21.1, pp. 421–432, the complier average causal effect)

Week 10: Regression discontinuity — the effect at the cutoff

- Problem set 4 distributed

Required readings

Theory

- Cattaneo et al. (2020, Ch. 2, pp. 8–19; Ch. 3, pp. 20–38; Ch. 4, pp. 39–87; Ch. 5, pp. 88–108)

Application

- Lee (2008)

Recommended readings

- Lee and Lemieux (2010)
- Hansen Vol II Ch. 21 (pp. 755–767)
- Caughey and Sekhon (2011)
- Eggers et al. (2015)
- Gelman et al. (2021, §21.3, pp. 432–440, fuzzy regression discontinuity as an IV problem)

Week 11: Difference-in-differences on panel data — the effect for the treated

Required readings

Theory (panel structure)

- Hansen Vol II §§17.1–17.13 (pp. 609–625), §§17.24–17.25 (p. 637)
- Beck and Katz (1995)

Theory (difference-in-differences)

- Hansen Vol II Ch. 18 (pp. 664–676)
- Roth et al. (2023)

Application

- Paglayan (2019)

Recommended readings

- Goodman-Bacon (2021)
- Callaway and Sant'Anna (2021)
- Ward (2020)
- Gelman et al. (2021, §21.4, pp. 442–445, two-group difference-in-differences and parallel trends; no staggered case)

Week 12: Grouped and censored data — multilevel and event history models

⋆ **Problem set 4 due**

Required readings

Theory (multilevel)

- Gelman and Hill (2007, Ch. 12, pp. 251–278)
- Bell et al. (2019)

Theory (event history)

- Hernán and Robins (2020, Ch. 17, pp. 227–240, causal survival analysis)
- Box-Steffensmeier and Jones (2004, Ch. 2, pp. 7–20, censoring and the hazard and survivor functions)

Application

- Barnes and Burchard (2013)

Recommended readings

- Gelman and Hill (2007, Ch. 11, pp. 237–247, fixed versus random effects; Ch. 13, pp. 279–283, varying slopes)
- Hansen Vol I Ch. 15 (pp. 303–312, shrinkage)
- Hansen Vol II §§17.5–17.6 (pp. 613–616, random effects)
- Hansen Vol II Ch. 27 (pp. 859–862, censoring and selection)
- Roback and Legler (2021, Chs. 8–9): a free alternative, available here
- Box-Steffensmeier and Jones (2004, Ch. 4, pp. 47–68, the Cox proportional hazards model and the risk set)
- Steenbergen and Jones (2002)
- Box-Steffensmeier and Zorn (2002)

Week 13: Spatial data — dependence among neighboring outcomes

⋆ **Draft of final paper due at the end of the week**

Required readings

Theory

- Ward and Gleditsch (2007, Ch. 1 §§3–7, pp. 7–28; Ch. 2, pp. 29–54; Ch. 3, pp. 55–64, especially §5, p. 58, spatially lagged y versus spatial errors)

Application

- Franzese and Hays (2008), reanalyzing Swank and Steinmo (2002)

Recommended readings

- Franzese and Hays (2007)
- Pebesma and Bivand (2023): the `sf` and `spdep` packages, available here

Week 14: Final paper presentations

⋆ **Final paper due the last day of finals week**

A 15-minute talk on your final paper, plus 5 minutes of discussant remarks on one classmate's paper; slides for both are due by the end of the day before, so the presentations run back-to-back without setup time and fill the whole session.

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